Ajay Rajendra Kumar

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Education

Northeastern University — **Boston, USA**
M.S. in Computer Science; GPA: 4.0/4.0 — *Sep. 2024 - May 2026*

Manipal Institute of Technology — **Manipal, India**
B.Tech. in Computer Science (Minor: Computational Intelligence); GPA: 9.3/10.0 — *Aug. 2020 - Jul. 2024*

Experience

Research Assistant — **Nov. 2025 – Present**
FarSight Lab, Northeastern University (Advisor: Prof. Lorenzo Torresani) — *Boston, MA*

- Improved video question answering performance of the PerceptionLM-8B by 12% on the EgoBlind benchmark through prompt optimization and supervised fine-tuning.
- Engineered an agentic pseudo chain-of-thought generation pipeline integrating SAM3, Qwen-2.5-VL-7B, and PerceptionLM-8B to automatically generate grounded visual concepts and textual evidence, augmenting the Open-o3-Video dataset with 10K+ high-quality training samples.
- Trained Qwen-2.5-VL-7B using a two-stage curriculum comprising LoRA-based supervised fine-tuning (SFT) followed by GRPO reinforcement learning (RL) on the synthesized grounded supervision dataset.
- Achieved a 5% improvement on the V-STaR video benchmark, with consistent gains across other benchmarks; currently extending the approach to long-horizon multimodal reasoning via audio-video evidence retrieval and memory-augmented reasoning.

Research Assistant — **Apr. 2025 - Mar. 2026**
Visual Intelligence Lab, Northeastern University (Advisor: Prof. Huaizu Jiang) — *Boston, MA*

- Designed and trained a lightweight dense feature matching network to estimate pixel-wise correspondence confidence, filtering unreliable matches to improve the robustness and accuracy of downstream camera pose estimation.
- Architected a scalable Kubric-based synthetic data generation pipeline to render 2K+ stereo video sequences with diverse camera trajectories, object configurations, and HDRI environments for training the confidence estimation network.
- Co-developed NeuSLAM, a dense visual SLAM system integrating the confidence-guided front-end with an efficient classical pose-tracking back-end, achieving state-of-the-art F1-all of 9.9% on KITTI 2015 optical flow and competitive stereo disparity on KITTI 2015 and ETH3D in zero-shot synthetic-to-real evaluation.
- Contributed to the deployment, optimization, and experimental validation of NeuSLAM on Jetson Orin Nano-based robotic platforms, achieving trajectory accuracy competitive with DROID-SLAM at 6.5× faster speed and 87% lower GPU memory on TUM RGB-D and FLSea-VI datasets; accepted at the CVPR 2026 Demo Track and the ICRA 2026 S2S Workshop.

Graduate Research Assistant — **Feb. 2025 - May 2026**
D'Amore-McKim School of Business, Northeastern University (Profs. Anna Lamin, Valentina Marano) — *Boston, MA*

- Engineered an end-to-end data acquisition pipeline to scrape and preprocess CSR disclosures from Indian companies, curating a dataset of 100K+ records spanning five years for downstream empirical analysis.
- Built scalable Python-based ETL pipelines for JSON and PDF ingestion, incorporating structure-aware document chunking, metadata extraction, and automated preprocessing to support retrieval and information extraction workflows.
- Architected an LLM-powered document intelligence pipeline combining semantic retrieval, targeted OCR, and OpenAI models to extract, clean, normalize, and transform unstructured reports into structured tabular datasets for downstream analytics.
- Scored 10K+ Crunchbase company profiles using a two-stage LLM pipeline (embedding retrieval + GPT-4.1 Likert scoring) against ten research criteria, enabling empirical analysis of domestic market orientation of Indian firms.

Software Development Intern — **Jun. 2023 – Aug. 2023**
Fidelity Investments — *Bengaluru, India*

- Reduced dashboard query latency by 90% by migrating the underlying data warehouse from Oracle to Snowflake and re-architecting query execution for large-scale change-ticket analytics.
- Delivered full-stack features for an internal change-ticket management platform using Angular and Spring Boot, redesigning core workflows for creating, tracking, and managing engineering change requests.
- Designed and integrated RESTful APIs using Spring Boot to connect the Angular frontend with Snowflake-backed backend services, enabling efficient ticket creation, tracking, and reporting workflows.

Publications

"NeuSLAM: Dense Visual SLAM on Edge Devices" – under review at *IEEE IROS 2026*; accepted to *ICRA 2026 S2S Workshop* and *CVPR 2026 Demo Track* (co-author)

"Addressing Vaccine Misinformation on Social Media by Leveraging Transformers and User Association Dynamics" – *ICMLDE 2023* (co-first author)

Technical Skills

Programming: Python, C/C++, Java, SQL, R
Frameworks & Tools: PyTorch, TensorFlow, vLLM, HuggingFace, PEFT (LoRA, QLoRA), bitsandbytes, Weights & Biases, Git
Distributed Training: PyTorch Distributed (DDP/FSDP), Slurm, DeepSpeed
Web & Data: Angular, Spring Boot, Snowflake, Oracle DB
Research Areas: Multimodal Learning, Long-Horizon Reasoning, Video Understanding, Visual SLAM, Deep Learning